Crime LA Project Data Analysis

PGP DSBA – SQL Guided Project Report

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1. Crime LA (Los Angeles) Project Data Analytics using SQL

Business Context

People throughout Los Angeles are concerned about recent reports of crimes in numerous locations. The mayor of Los Angeles has established a new Criminal Investigation Division to study how and why crime is on the rise and the elements that contribute to it so that officials may take the necessary steps to keep the city's residents safe.

Objective

As a member of the Analytics Division, you are aware that numerous questions must be answered utilizing CID data. Import the dump file that contains various tables that are present in the database. Using the data, react to the questions and write a detailed report for the authorities to utilize in taking action against crimes in Los Angeles.

2. Data Description

The data contains the different attributes of leads and their interaction details with Crime LA (Los Angeles) Project Data Analytics using SQL. The detailed data dictionary is given below.

2.1 Crimes Data Dictionary

- report_no int Unique Record number
- incident_time time Time of the crime occurrence
- complaint_type object Which mode complaint is registered
- cctv_flag object At the crime scene whether the CCTV footage is available or not
- precinct_code int Unique code representing a precinct in a city
- area_code int Unique code of that locality
- population_density int Population density of the locality
- area_name object Locality name
- cctv_count int Number of CCTV active in the locality
- rounds_per_day int How many rounds patrolling cars are taking for each area code.
- crime_code int Unique code for the crime type
- crime_type object Description of the crime committed
- weapon_code int Unique code for the weapon used
- weapon_desc object Type of weapon used for the crime
- case_status_code object Unique code for the case status(IC- Investigation Continues, AA- Adult Arrest, AO- Adult Other)
- case_status_desc object Defines the status code provided
- victim_code int Unique code of the victim
- victim_name object Name of the victim
- victim_age int Victim age

- victim_sex object Gender of the victim
- was_victim_alone object Was victim alone during crime scene
- is_victim_insured object Is the victim insured
- offender_code int Unique code of the offender
- offender_name object Name of the offender
- offender_sex object Gender of the offender
- offender_age int Age of the offender
- repeated_offender object Is the offender repeated the crime
- no_of_offences int How many cases are registered against that offender
- offender_relation object Is the offender is a relative of the victim
- officer_code int Unique code of the officer
- officer_name object Name of the officer
- officer_sex object Gender of the officer
- avg_close_days int Average days officer is taking to solve a case
- week_number int In which week the crime occurred

3. Business Questions

3.1 Question 1: What is the total number of crimes for each crime status?

Output:

- The screenshot shows 3 rows of the output which means the Count of the crimes for different case statuses.
- The screenshot contains the "SQL Query Passed" (Query 1) message and output table.

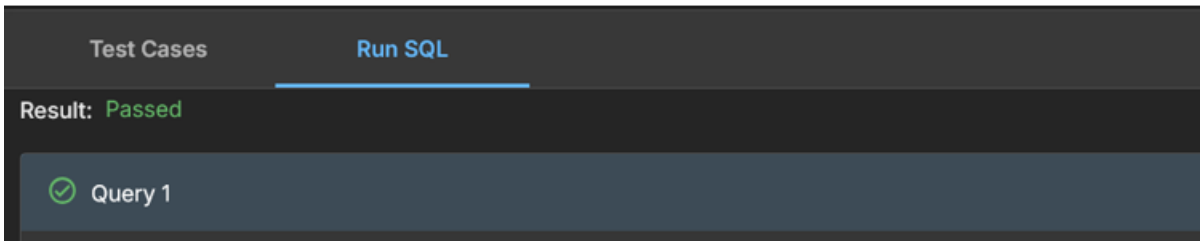


Table 1 Query1 Passed

Output:

Showing 3 rows

case_status_desc	case_count
Adult Arrest	94
Adult Other	38
Invest Cont	1186

Table 2 Query1 Output table

Observations and Insights: -

- The total number of crimes is 1318 for all the case statuses.
- Invest Cont - Case status has the highest number of crimes (1186).
- Adult Other - Case status has the lowest number of crimes (94).
- Adult Arrest – Case status has the 38 number of crimes
- There are many cases under the Investigation Continues which will reduce crimes and once completed it will increase the Adult arrest count & Adult Other count.

3.2 Question 2: Which was the most frequent crime committed each week?

Output:

- Ensure the screenshot contains the "SQL Query Passed" (Query 2) message and the output table.
- The screenshot shows 4 rows of the output which means the Count of the number of crimes for different case statuses.

Test Cases Run SQL

Result: Passed

Query 1

Query 2

Table 3 Query 2 Passed

Output:

Showing 4 rows

week_number	crime_type	crimes_reported
1	BURGLARY FROM VEHI...	24
2	BURGLARY FROM VEHI...	40
3	BATTERY - SIMPLE AS...	48
4	BURGLARY FROM VEHI...	59

Table 4 Query 2 Output table

Observations and Insights:

- Number of crimes reported each week and crime type.
- These are the highest crimes committed each week so need to increase the security system to reduce the crime rate.

3.3 Question 3: Does the existence of CCTV cameras deter crimes from happening?

Output:

- Ensure the screenshot contains the "SQL Query Passed" (Query 3) message and the output table.
- The screenshot shows 10 rows out of 11 rows of the output which means the existence of CCTV cameras deter crimes from happening.

Test Cases Run SQL

Result: Passed

Query 1

Query 2

Query 3

Table 5 Query 5 Passed

Output:

Showing first 10 rows out of 11 rows

area_name	cctv_count	cases_reported
Hollywood	280	122
Newton	275	74
West Valley	268	156
Northeast	255	45
Van Nuys	250	158
West LA	244	156
Harbor	180	71
Hollenbeck	170	189
Southwest	168	56
Rampart	165	233

Table 6 Query 3 Output table

Observations and Insights:

- The Rampart area has reported 233 cases, and the CCTV count is 165. Due to the lower number of CCTVs, more cases are reported, so it is better to increase CCTV to control the crime rate.
- The area name Northeast has reported 45 cases and fixed CCTV 255 count. Northeast has reported lower number of cases due to more CCTV's so raise in CCTV's will reduce the cases reported.

3.4 Question 4: How much footage has been recovered from the CCTV at the crime scene?

Output:

- Ensure the screenshot contains the "SQL Query Passed" (Query 4) message and the output table.
- The screenshot shows 1 row out of the output which means the footage has been recovered from the CCTV at the crime scene.

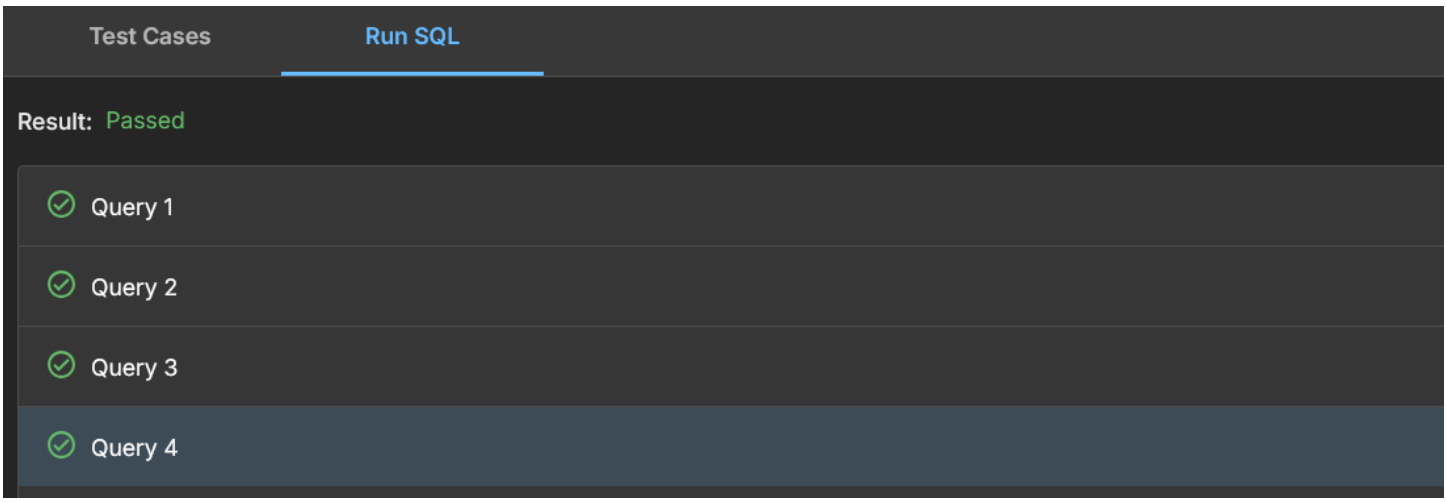
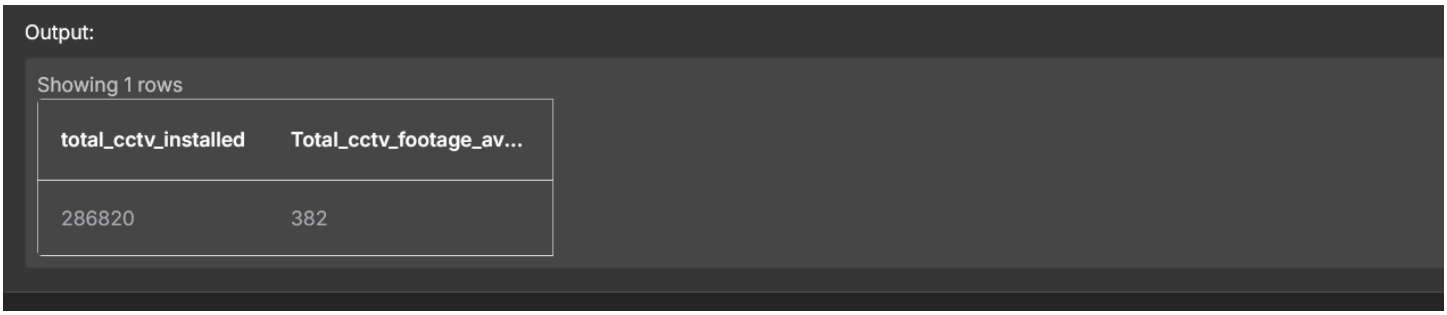


Table 7 Query 4 Passed



total_cctv_installed	Total_cctv_footage_av...
286820	382

Table 8 Query 4 Output table

Observations and Insights:

- The total CCTV's installed are 286820 and Total CCTV footage available 382.
- CCTV footage is available in 382 out of all the cameras at the crime scenes which causes to missing crime footage. Hence, it needs to maintain the footage available to find the offenders.

3.5 Question 5: What is the frequency of various complaint types?

Output:

- Ensure the screenshot contains the "SQL Query Passed" (Query 5) message and the output table.
- The screenshot shows 3 rows of the output which means the frequency of various complaint types.

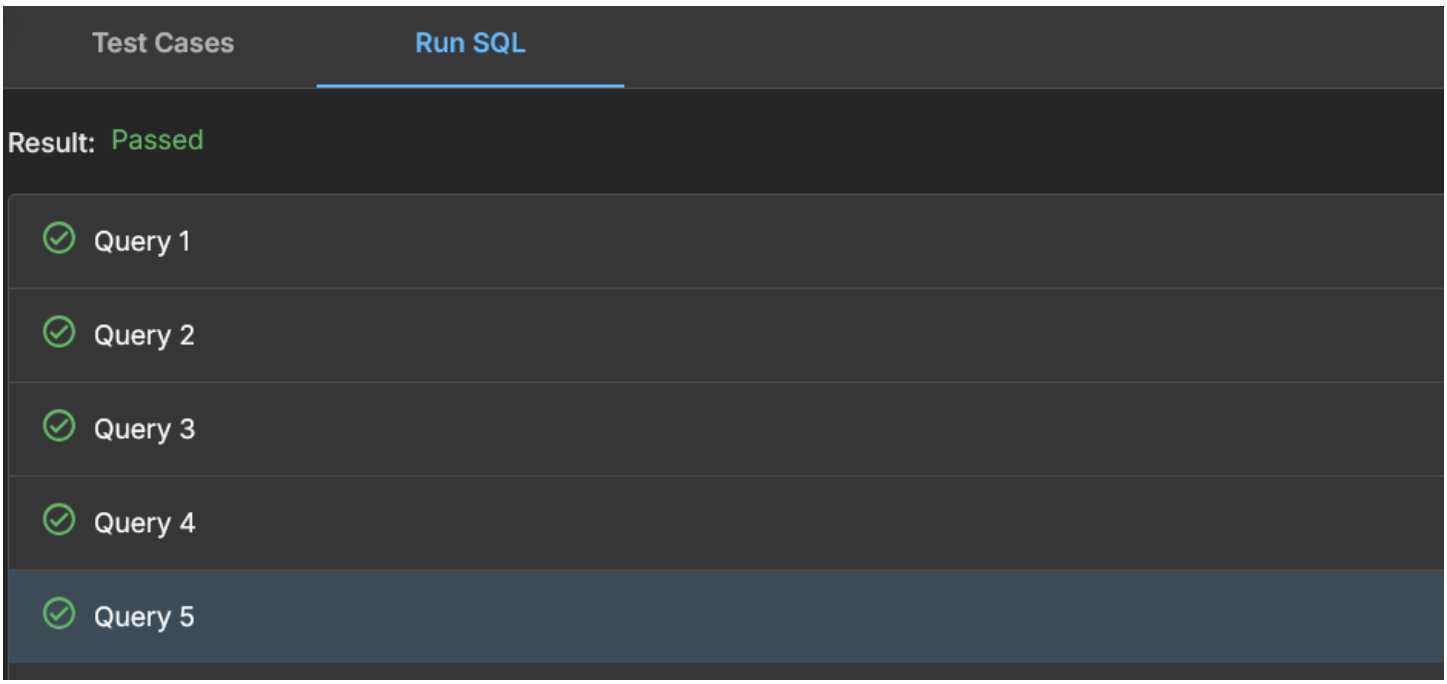
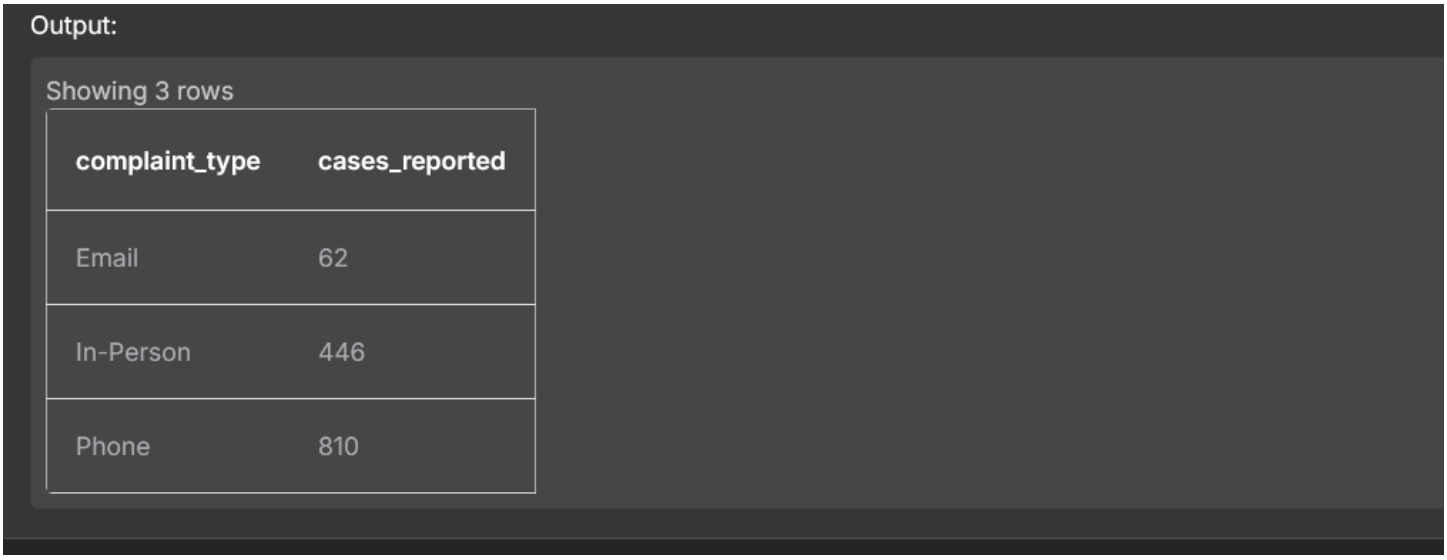


Table 9 Query 5 Passed



complaint_type	cases_reported
Email	62
In-Person	446
Phone	810

Table 10 Query 5 Output table

Observations and Insights:

- The frequency of complaint type Phone has 810 cases which is the highest one.
- The frequency of In-Person complaint type has 446 cases reported which is the highest one.
- The frequency of complaint type Phone has 62 cases which is the least.
- Insights reflect the victims have been informing the complaints over the phone has more frequency than the In-Person & Emails.

3.6 Question 6: Is crime more likely to be committed by the relation of victims or

Output:

- Ensure the screenshot contains the “SQL Query Passed” (Query 6) message and the output table.

- The screenshot shows 2 rows of the output which means the crime is more likely to be committed by the relation of victims or output.

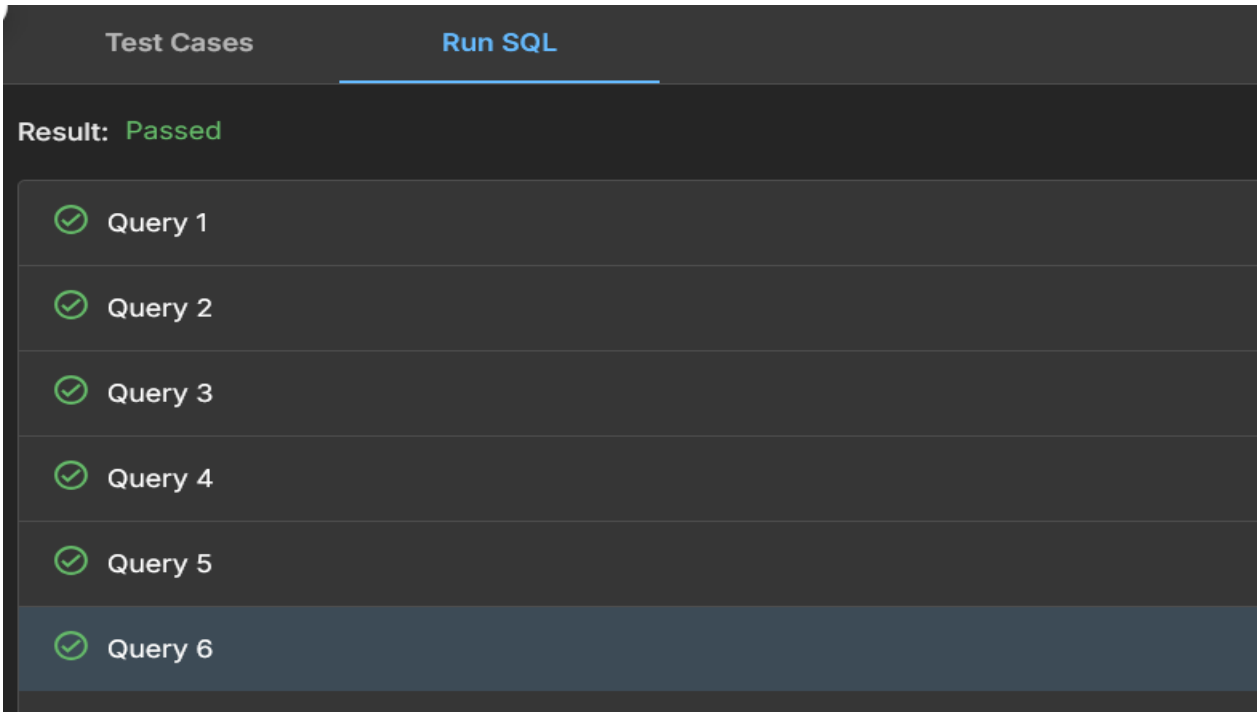
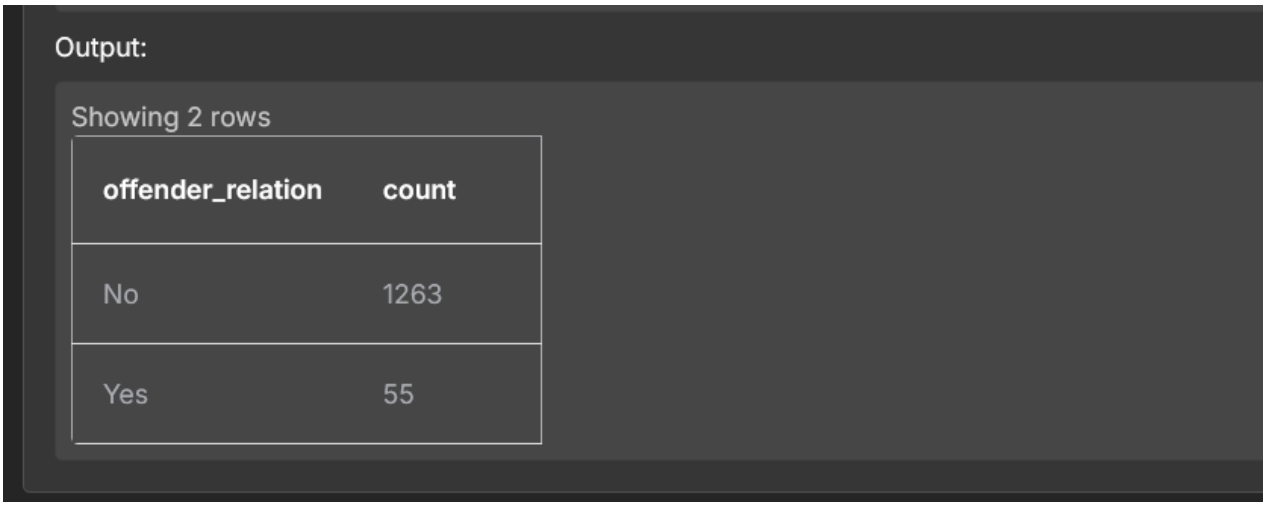


Table 11 Query 6 Passed



offender_relation	count
No	1263
Yes	55

Table 12 Query 6 Output table

Observations and Insights:

- As per the output table Offender relation count Yes is 55 and No relation is 1263.
- Crime is more likely to be committed by the no relation of victims or Output is higher than the offender relation count with victims. So crime is not dependent on the relation of the victims.

3.7 Question 7: Is crime more prevalent in areas with a higher population density, fewer police personnel, and a larger precinct area?

Output:

- The screenshot shows the first 7 rows of the output where the crime more prevalent in areas with a higher population density, count the total areas, total officers, and cases reported in each precinct code.
- The screenshot contains the "SQL Query Passed" (Query 7) message and output table.

Table 13 Query 7 Passed

Showing 7 rows

precinct_code	pop_density	total_areas	total_officers	cases_reported
1	616200	156	156	156
2	1136862	159	159	159
3	1513542	314	314	314
4	1980500	233	233	233
5	713766	145	145	145
6	1263654	189	189	189
7	390400	122	122	122

Table 14 Query 7 Output table

Observations and Insights:

- Precinct code 4 has a higher population density of 1980500 with 233 police personnel, 233 of the precinct area also reported 233 cases.
- There are 122 police personnel allotted for 390400 population density with a precinct area of 122.
- The precinct code 3 of the larger precinct area is 314 which has 314 cases reported.
- Insights reflect insufficient officers to the respective population density in area which causes the crimes. More crimes in the place of more population density.

3.8 Question 8: At what parts of the day is the crime rate at its peak? Group this by the type of crime. Use the following mapping to divide the day into five parts.

00:00 to 05:00 = Midnight,
05:01 to 12:00 = Morning,
12:01 to 18:00 = Afternoon,
18:01 to 21:00 = Evening,
21:00 to 24:00 = Night

Output:

- The screenshot shows the first 10 rows out of 164 rows of the output which means at what parts of the day is the crime rate at its peak.
- The screenshot contains the "SQL Query Passed" (Query 8) message and output table.

Table 15 Query 8 Passed

Test Cases | Run SQL

Output:

Showing first 10 rows out of 164 rows

dayparts	crime_type	crimes_reported
Afternoon	ARSON	2
Afternoon	ASSAULT WITH DEADL...	15
Afternoon	ATTEMPTED ROBBERY	2
Afternoon	BATTERY - SIMPLE AS...	54
Afternoon	BATTERY POLICE (SIM...	2
Afternoon	BATTERY WITH SEXUA...	8
Afternoon	BIKE - STOLEN	12
Afternoon	BRANDISH WEAPON	3
Afternoon	BUNCO, ATTEMPT	1
Afternoon	BUNCO, GRAND THEFT	2

Table 16 Query 8 Output table

Observations and Insights:

- The crime type BATTERY - SIMPLE ASSAULT has 54 crimes reported in the afternoon (between 12:00 and 18:00).
- The crime type BUNCO, ATTEMPT has 1 or One crime reported in the afternoon (between 12:00 and 18:00).
- Insights reflect the short-term actions or short temper where the victims making the intentional harmful actions of another person.

3.9 Question 9: At what point in the day do most crimes occur in different localities? Use the same mapping provided in Question 8 to divide the day into five parts.

Output:

- The screenshot shows the first 10 rows out of 11 rows of the output which means at what point in the day more crimes occur in a different locality.
- The screenshot contains the "SQL Query Passed" (Query 9) message and output table.

Table 17 Query 9 Passed

Output:

Showing first 10 rows out of 11 rows

area_name	dayparts	cases_reported
77th Street	Afternoon	17
Harbor	Afternoon	23
Hollenbeck	Afternoon	55
Hollywood	Afternoon	40
Newton	Afternoon	21
Northeast	Afternoon	16
Rampart	Afternoon	77
Southwest	Afternoon	19
Van Nuys	Afternoon	47
West LA	Afternoon	42

Table 18 Query 9 Output table

Observations and Insights:

- 77 crimes occurred in Rampart locality in the afternoon and it has reported the highest rank.
- 16 cases reported in the area name Northeast in the afternoon which is the lowest one.

3.10 Question 10: Which age group is more likely to fall victim to crimes at certain points in the day? Use the same mapping provided in Question 8 to divide the day into five parts. Additionally, use the following mapping to divide the age group.

Age 0 to 12: kids
13 to 23: teenage
24 to 35: middle age
36 to 55: adults
56 to 120: old

Output:

- The screenshot shows the first 10 rows out of 25 rows of the output which means the Count of the crimes for different case statuses.
- The screenshot contains the “SQL Query Passed” (Query 10) message and output table.

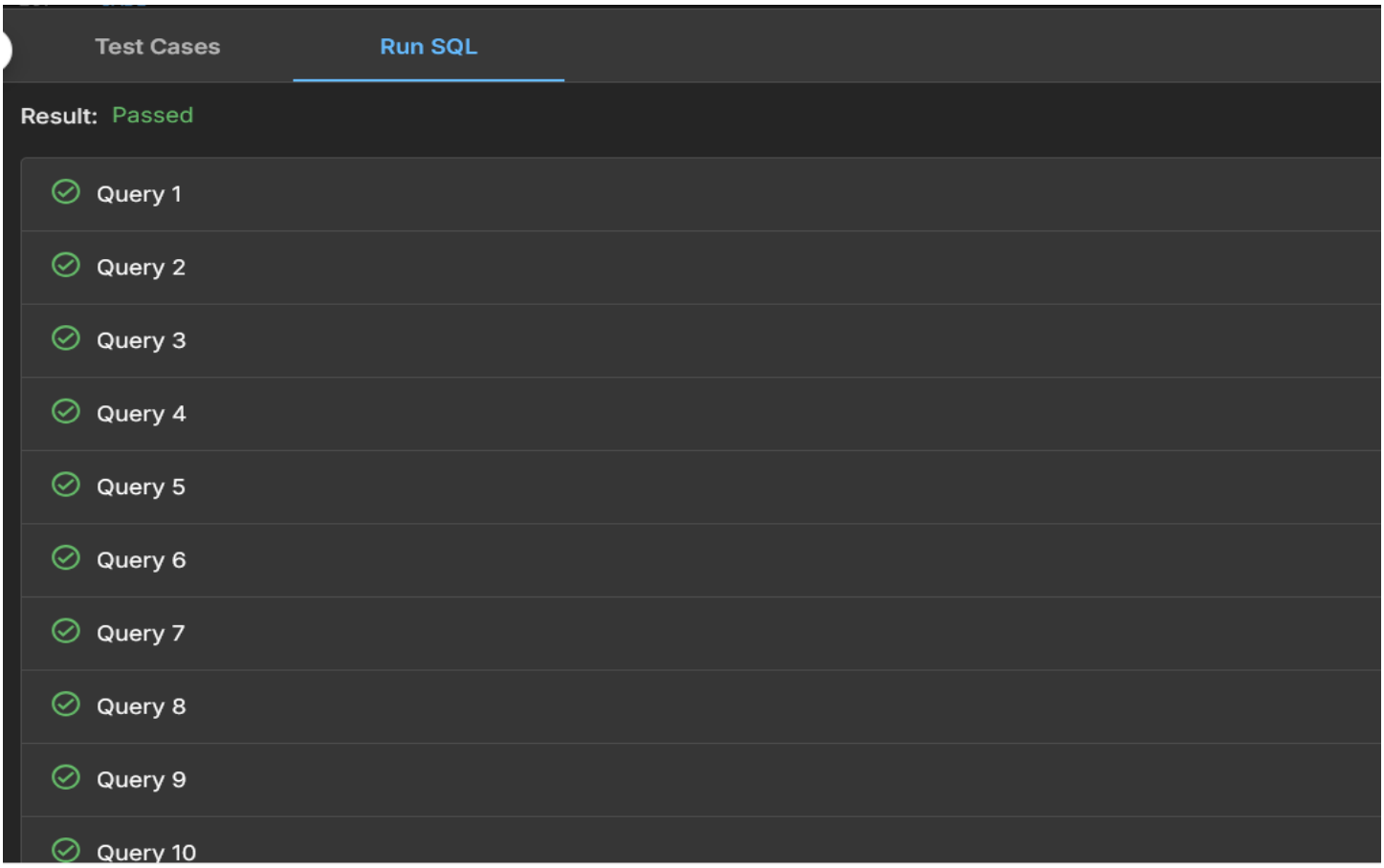


Table 19 Query 10 Passed

Output:

Showing first 10 rows out of 25 rows

dayparts	age_cat	cases_reported
Afternoon	Adults	229
Evening	Adults	183
Morning	Adults	163
Midnight	Adults	110
Afternoon	Middle age	74
Afternoon	Old	67

Morning	Middle age	64
Evening	Middle age	60
Night	Adults	57
Evening	Old	53

Table 20 Query 10 Output table

Observations and Insights:

- This case counts the number of crimes grouped by the age group and part of the day.
- Afternoon time under the adult category more cases (229) were reported. Increasing security controls for adults in the afternoon time to reduce the crime rate in Los Angeles.
- Old age group of people (53) are more likely to fall victim to crimes at certain points in the evening time. It has to increase CCTV to save the old age people from the crimes at evening time.

4. Crime Metrics Overview

No. of Precinct	Total Crimes Reported	No. of Areas	No. of Offenders
54	3063	11	1318
No. of Officers	**Total Population**	**No. of CCTV's**	**No. of Cases where Status is IC**
1318	62683	2405	1186

5. Recommendations

As a member of the Analytics Division notifying the below recommendations for the authorities to utilize in taking action against crimes in Los Angeles.

- The total number of crimes under Investigation Continues are 1186 waiting for resolution. Once these in-process crimes resolved the crime rate will be reduced.
- CCTV's installed but very less footage has recovered so it needs to increase footage recover points.
- Many of the victims are using phone to complain about the crime so it officers educate the victims about alternative options like email & In-Person.
- Crime more likely to be committed by the relation of victims or Output is less than crime committed by the no relation of victims. Hence, victims should be careful about the crimes irrespective of the relationships.
- More cases reported in more population density places means there is a chance of occurring more crimes so need to increase the police personnel and CCTV's footage recovery to control the crime and improve piece in the public.
- Afternoon part of the day has more crimes due to ASSAULT so educate victims/offenders in the society as it is harmful to the society and leads to raise criminal background. Therefore, Offenders may realize and victims stay without any fear. So afternoon security controls should fix.
- Adults on afternoon reported more cases so need to train them how to safeguard themselves from the offenders.